

MOHAMMED AFIL FARHAN

AI & ML SPECIALIST . DATA ENGINEER

afilfarhan0008@gmail.com , [+966 557337857](tel:+966557337857) , Riyadh, Saudi Arabia

SUMMARY

AI Engineer and Machine Learning specialist with a Master's in Artificial Intelligence from Heriot-Watt University, experienced in building production-grade multi-agent systems, RAG pipelines, and full-stack AI applications. Proficient in Python and JavaScript/TypeScript, with a track record of shipping expandable, user-centric automation solutions across startups and enterprise environments, leveraging tools like LangChain and REST APIs.

EDUCATION

Heriot-Watt University, United Kingdom

2022 - 2023

MSc Artificial Intelligence

Graduated with Merit

Srinivas University, India

2019 - 2022

BCA - Cloud Technology & Information Security

8.6 / 10 CGPA

SKILLS AND CERTIFICATIONS

Technical Skills:

Programming Languages: Python, Java (OOP), C, Kotlin, Golang, Julia, JavaScript, TypeScript, PHP

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, NLTK, OpenCV, Google Mediapipe, LangChain, Keras, LlamaIndex, LangGraph, LiteLLM, NLP, Computer Vision, LLMs, Deep Learning, Retrieval Augmented Generation (RAG), Prompt Engineering, Generative AI, MLOps, LangSmith

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, CloudFormation, CI/CD Pipelines

Databases: MySQL, SQL, MongoDB, Supabase, PostgreSQL, Pinecone, ChromaDB, Vector Databases

Tools & IDEs: Git, GitHub, GitLab, JUnit, VS Code, IntelliJ, Eclipse, Anaconda, Cursor, Zapier, n8n, Claude code, ORC, Opencode, MCP's

Web Technologies: HTML5, CSS3, FastAPI, Node.js, Bootstrap, React, Express, Flask, Django, REST APIs, Vercel, Firebase

Data Engineering / Big Data: NumPy, Pandas, Redis, Apache Kafka, Google BigQuery, PySpark, Airbyte

Additional Skills: JSON

Certifications:

Machine Learning Applications, Data Visualization using Python

EXPERIENCE

Data Engineer

January 2025 - June 2026

Bejoc Shipping L.L.C

Dammam, KSA

- Designed and maintained expandable ETL/ELT pipelines to ingest, transform, and load data from multiple sources into centralized data warehouses and data lakes, ensuring reliable and high-performance data movement across the organization
- Engineered data quality control frameworks including validation, cleaning, and automated testing processes to enforce data integrity, governance protocols, and compliance standards (GDPR) across all storage and pipeline stages.
- Automated data collection, processing workflows, and system integrations through infrastructure scripting, reducing manual overhead and improving pipeline reliability and operational efficiency.
- Optimized database queries and data pipelines for performance and cost-efficiency, partnering cross-functionally with data scientists, analysts, and software engineers to deliver clean, structured datasets aligned with business requirements

AI Engineer

September 2023 - December 2024

Optima Partners

Scotland, United Kingdom

- Engineered multi-agent AI systems leveraging LLMs and LangChain to automate complex workflows, collaborating cross-functionally with ML teams to integrate AI capabilities directly into core product features, demonstrating expertise in agentic AI and automation.
- Designed and deployed Retrieval-Augmented Generation (RAG) pipelines using vector databases and embedding models to enable accurate, context-aware AI responses at scale.
- Led end-to-end AI implementation across the product lifecycle, from model selection and prompt engineering to evaluation and production deployment, aligned with emerging LLM best practices and demonstrating expertise in LLM prompting.
- Owned full-stack AI/ML engineering responsibilities, building, fine-tuning, and optimizing models and AI-powered features end-to-end, driving measurable improvements in system performance, retrieval accuracy, and output quality.

Web Development Intern

April 2022 - June 2022

Yenapoya Incubator

India

- Designed and implemented front-end UI components for 3 internal projects (hospital, taxi, and canteen websites) using Bootstrap and jQuery, enhancing user experience and interface consistency.
- Optimised healthcare database performance by over 15% through advanced SQL query restructuring measured using SSMS.
- Developed 2 automated dynamic web PDFs for hospital and taxi websites using TCPDF in PHP, streamlining data entry and printing processes.

PROJECTS

Socially Interactive Conversational AI with Expressive Emotions using LLMs (Master's Thesis)

- Developed a multimodal conversational AI system for the Furhat social robot, integrating Large Language Models (GPT-3.5) with the Furhat SDK to generate context-aware dialogue and synchronized emotional facial gestures for human-robot interaction.
- Designed and implemented prompt engineering and few-shot learning pipelines to improve gesture prediction and conversational quality, optimizing LLM responses while reducing prompt costs and maintaining high contextual relevance.
- Built the application using Kotlin, Python, OpenAI API, and Furhat SDK, integrating ASR, NLU, TTS, speech recognition, dialogue management, and custom gesture generation into a modular conversational architecture.
- Evaluated system performance through Word Error Rate (WER) analysis and user studies with human participants, assessing conversational accuracy, multimodal interaction quality, and the impact of emotional gestures on user engagement and satisfaction.

Academic AI Research & Engineering Projects

- Developed customised neural network architectures and hyperparameter optimisation frameworks using Python. Utilised meticulous feature engineering techniques to ensure data cleanliness and maintained clean, reusable code.
- Formulated and implemented genetic algorithms and particle swarm optimisation techniques to efficiently solve benchmark problems.
- Led end-to-end development of a medical e-commerce platform with an integrated NLP-powered chatbot built on Flask, NLTK, and PyTorch, delivering a full-stack AI solution with conversational capability and seamless user experience.
- Implemented a range of ML algorithms covering supervised and unsupervised learning, including CNNs, RNNs, and GANs, for specific project needs. Applied reinforcement learning strategies to optimise model performance across diverse applications.
- Built a real-time American Sign Language (ASL) recognition app that runs entirely in the browser, no backend required, using MediaPipe for hand and body tracking and ONNX Runtime for on-device model inference, with a Transformer-based model capable of detecting both individual signs and full sentences from live webcam input.
- Designed and deployed a production-grade RAG chatbot for a digital commerce marketplace using LangChain, a ReAct agent architecture, and Redis caching, incorporating anti-hallucination guardrails and context-aware retrieval to support real-time product discovery, showcasing expertise in building reliable AI agents with LangChain.